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Personal Protective Equipment (PPE)

1.0 BACKGROUND

The extensive use of PPE in the semiconductor industry has been driven traditionally from the desire to protect the fab employee from the numerous chemical and physical agents in the fab environment. As discussed in Ch. 4: Ventilation, another equally compelling goal is to isolate the worker from the wafer for process cleanliness reasons. The standard cleanroom gowning and PPE protocols require extensive use of cleanroom clothing, head and foot coverings, “cleanroom” gloves, face coverings, and safety glasses. Figure 5.1 illustrates a typical fab worker in cleanroom garments. Cleanroom gloves are typically made of either vinyl, PVC, or natural rubber. The intended purpose of these gloves is entirely for particle isolation, but they also can act as a simple physical barrier for toxic metal residues that may be present in some areas of the fab environment. However, when PPE is needed for safety reasons, such as to protect the skin from chemical contact, additional safety-specific PPE is necessary such as chemical protective gloves, safety glasses, aprons, arm guards, etc.

The primary focus of this chapter is on two specific areas of PPE—chemical protective (CP) gloves and respiratory protection. In both of these areas, the focus is to provide insights into the semiconductor fab environment and relevant information that is considered significant by the authors (rather than a general overview of these two topics).



Figure 5.1. Typical fab worker in cleanroom garments.^[1] (Van Zant, P., *Microchip Fabrication*, © 1990, McGraw-Hill, reprinted with permission.)

2.0 CHEMICAL PROTECTIVE GLOVES

Contact with the chemicals used in semiconductor fabrication processing has historically accounted for a significant portion of all reported illnesses and injuries in the semiconductor industry.^[2] The extensive use of many categories of hazardous materials such as acids (with special emphasis on hydrofluoric acid—HF), bases, organic solvents (including the glycol ethers), toxins (such as arsenic, antimony, and lead), and materials with unique chemical properties such as n-methyl-2-pyrrolidone (M-Pyrol, or NMP) and hexamethyldisilazane (HMDS). All of these materials present the potential for a dermal exposure hazard. While many of the fabrication processing steps are automated and isolated in a state-of-the-art Class 0.1 fab area, there are many times when the manual handling of liquids, solids, or other residues is still required. In the older Class 10 and 100 fabrication areas, chemicals may still be manually handled by fab area workers, and dispensing performed from chemical containers versus the state-of-the-art bulk chemical dispense systems. Some examples of manual chemical handling that present a hazard for direct contact include: routine maintenance activities; cleaning of equipment or work surfaces; material transfers; the cleanup of spills, leaks, or residuals in equipment or dispense

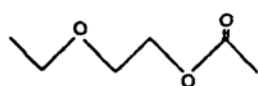
vessels; the final handling of hazardous waste/containers; and decontamination/decommissioning of process or support equipment surfaces.

With most of these tasks, the concern is short-term accidental contact and splashes rather than long-term immersion of the gloved hand in a chemical bath. This, coupled with the advantages of keeping the glove selection process simple, makes it desirable to limit glove selection for routine operations to only a few choices, e.g., one acid latex glove for acid use and one solvent-resistant glove for solvent operations. Certain exceptions are then made for operations where the nature of the exposure or the chemistries used necessitate a different glove.

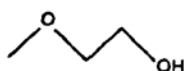
2.1 Glycol-Ether Derivatives

For certain solvents used in semiconductor processing, dermal exposure is of particular concern because they are readily absorbed through intact skin. Included among these are several ethylene glycol ether derivatives (e.g., ethylene glycol monoethyl ether acetate) which are used in the photolithographic resist formulations and are recognized animal reproductive toxins. A series of experiments were performed by Zellers, et al., to determine the effectiveness of different gloves with glycol ether mixtures and pure components that are used in semiconductor processing.^[3] The glycol ether derivatives that were tested were: 2-ethoxyethyl acetate (2-EEA or Cellosolve® Acetate), n-butyl acetate (N-BA), 2-methoxyethanol (2-ME), 2-methoxyethyl acetate (2-MEA), and xylene. The glove materials that were tested were: butyl, nitrile, and natural rubber, as well as natural rubber/neoprene blend and a natural rubber/neoprene/nitrile blend. *Breakthrough times* (BT) and *permeation rates* (PR) were determined using a permeation cell according to the ASTM F739-85 method.^[4] It was concluded from the testing that butyl gloves showed by far the best BT and PR for the materials tested, with only the xylene or xylene/2-ME mixture showing a noticeable decrement. Butyl gloves were also found to be the least affected by the temperature changes used in the testing (25 and 37°C). Nitrile gloves were found to be the second best choice. It should be noted that this article's authors did conclude that the large differences in glove properties when exposed to mixtures or individual compounds suggest that empirical evaluation is the only means for determining the safe choice of protective materials. Mention was also made of dexterity problems with butyl gloves.

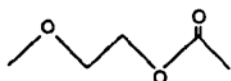
Zellers also did an additional study of the similar semiconductor photolithographic materials, with the addition of the propylene-glycol-derivative, propylene glycol monomethyl ether acetate (PGMEA) and N-methylpyrrolidone (NMP).^[5] PGMEA has been widely used as a substitute carrier solvent in photoresists to replace the ethylene glycol ether derivatives that have been recognized as animal reproductive toxins. Figure 5.2 identifies the structures of glycol-ether derivatives and N-methylpyrrolidone (NMP).^[5] Although there were numerous test results that were conflicting, the authors still found the butyl gloves to have the best BT and PR values, with the nitriles being the second choice. There was a notable exception for butyl gloves with high xylene concentrations (80%, as would be found in standard negative photoresist formulations), there was a significant alteration in the PR for 2-ME. The authors concluded that “the selection and use of chemically protective gloves must be based on detailed, task-specific evaluations of key variables such as duration of potential exposure, temperature, glove reuse practices, and the composition of solvent mixtures.”^[5]



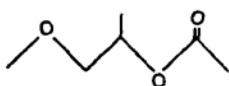
2-Ethoxyethyl Acetate(2-EEA)



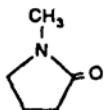
2-Methoxyethanol (2-ME)



2-Methoxyethanol Acetate (2-MEA)



1-Methoxy-2-propyl Acetate (PGMEA)



N-Methylpyrrolidone (NMP)

Figure 5.2. The structures of glycol-ether derivatives and N-methylpyrrolidone (NMP).^[5]
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2.2 N-Methyl-2-Pyrrolidone (NMP/M-Pyrol)

N-methyl-2-pyrrolidone (commonly referred to as NMP or M-Pyrol) which is used as a degreaser, a resist stripper, and as a solvent for phenolic die coat resins is known to cause skin irritation and to rapidly permeate through intact skin. The U.S. Environmental Protection Agency (EPA) in a final assessment document on NMP, found this material as a “significant risk” of reproductive and developmental effects for people working with NMP-based paint strippers.^[6] Obviously, the exposure potential from the use of NMP in paint stripping is completely different from its use in semiconductor processing as a photoresist stripper. In fab use, NMP is isolated from the fab worker in processing equipment that is a closed system. The low vapor pressure of NMP (0.342 mm Hg at 25°C) limits the saturated vapor concentration to about 450 ppm at room temperature. It has been reported that hydrolysis in air at relative humidities of 40–60% (a typical level found in a semiconductor fab area) can reduce maximum concentrations to approximately 130 ppm.^[7] The potential inhalation exposure hazard is low, but NMP is readily absorbed through the skin, and repeated or prolonged skin contact with the liquid can cause severe dermatitis.^[7] Another recent study of exposure to NMP vapors in semiconductor fab production facilities, symptoms of eye irritation and headache were observed at relatively low vapor concentrations (0.72–15 ppm). Based on these results, the authors of the study recommended a maximum exposure limit of 0.1 ppm, with a “skin” designation recommended in recognition of the dermal absorption potential of NMP.^[8] Dermal absorption may in fact represent the most important route of exposure to NMP. In research by Zellers and Sulewski,^[7] their study investigated the permeation of NMP through butyl and natural rubber gloves at various elevated temperatures (up to 50°C, with extrapolation to 70 and 93°C). They concluded that butyl-rubber gloves provided excellent protection under all test conditions, suggesting that these gloves be used for protection from NMP in all cases where particulate contamination from the butyl gloves can be tolerated. Unfortunately, butyl gloves are a known source of particulates and therefore may be difficult to get approved for use in a state-of-the-art fabrication area.

2.3 Hydrofluoric Acid (HF)

The use of hydrofluoric acid as an etchant in semiconductor processing is essential. HF’s ability to etch glass (SiO_2) was one of the key processing steps that allowed the precise removal of openings in a glass layer that had been applied to the exterior surface of the wafer through oxidation or epitaxial deposition. This use as an etchant is still a cornerstone

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of current IC processing. HF is typically used in a “straight” 49% solution as a liquid wet etchant at elevated temperatures or in a mixture known as *buffered oxide etch (BOE)* that contains diluted HF and ammonium fluoride. This process is usually performed in a wet sink that is equipped with LEV and splash shields. Fabrication workers wearing PPE then etch cassettes of wafers in a hot HF etch bath for a specified time period. This technique is essentially a total immersion process and the cassette is removed and quickly placed in an adjacent rinse bath containing ultrapure *de-ionized water (DI water)* that dilutes the HF and stops the etching process. The chemical etch baths are typically manually poured with the etch solution or the BOE mixture into the bath. Both the etching process and manual addition of etchants can cause splashing, spillage, or condensation of HF vapors on cooler surfaces. HF looks identical to water on a surface, and if the fab workers are not properly trained as to this fact, and performing excellent housekeeping of liquid residues, skin contact is almost inevitable.

Until the late 1980s, HF acid burns were a frequent occurrence in most production semiconductor fabrication areas. With an emphasis on training employees about the hazards of HF, better work practices in fab areas, the increased use of bulk chemical dispense systems, and tighter IC geometry requirements mandating the use of dry chemical etching (plasma etching), the incidence of HF burns has dropped significantly. HF is well respected in the semiconductor industrial hygiene community not due to the particular strength of the acid (it is generally considered to be a weak acid at the percentage solutions that are used), but partly because of this property: HF burns from dilute solutions do not cause immediate pain, and therefore no warning signal is provided to the exposed individual indicating the chemical is on the skin. This can result in prolonged exposure prior to recognition. The severity of the HF burn is a function of the duration of contact, the strength of the material, quality of the skin involved, and other physical factors. For chemicals like HF that have the ability to penetrate intact skin, the duration of contact is extremely important in determining the depth of the injury. HF is also unusual in that the uncharged form of HF is able to penetrate the lipid barrier and enter the deeper layers of the skin and the subcutaneous tissues. Once there, the fluoride ions exert their effect as an extremely corrosive material, and binds to the tissue surfaces. The ion attacks enzymes as well as the structural components of the cell membranes and organelles. It is this effect that is thought to cause the damage to the tissues rather than the effect of the hydrogen ion acting as an acid.^[9] Figure 5.3 graphically illustrates the appearance of a fingertip burn from contact with a 10% hydrofluoric acid without adequate treatment. Photo A is after 18 hours and Photo B is after 24 hours.^[9]

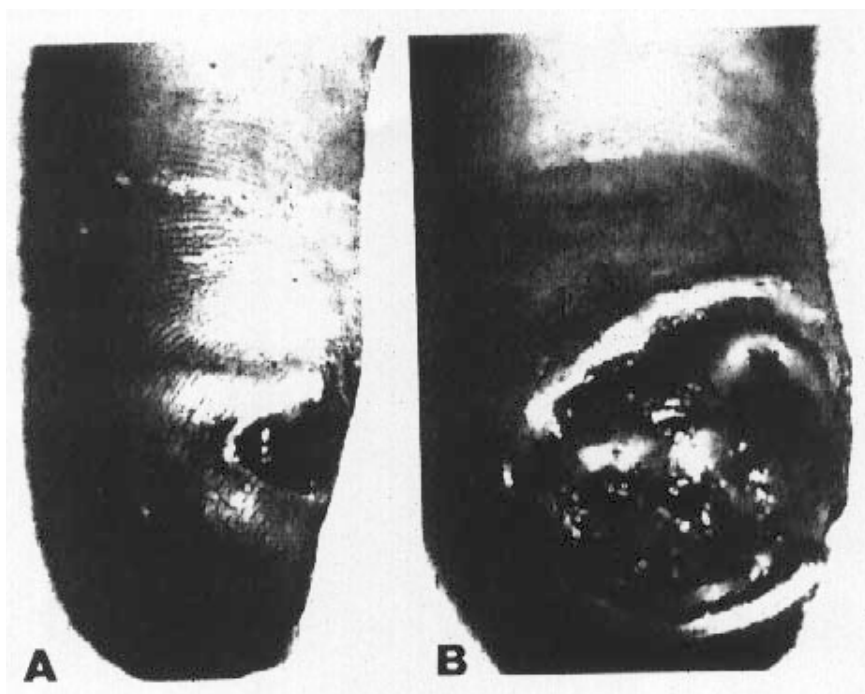


Figure 5.3. The appearance of a fingertip burn from contact with a 10% hydrofluoric acid without adequate treatment. Photo A is after 18 hours and Photo B is after 24 hours.^[9] (© 1986, Hanley & Belfus, reprinted with permission.)

The mainstay of treatment of chemical burns from hydrofluoric acid is the immediate dilution and decontamination of the affected (exposed) tissue. It is important for the worker to understand that the chemical continues to burn until it has fully reacted with tissue components, releasing all of its thermal energy or causing maximal poisoning to the cells through cytotoxic reaction. After immediately rinsing the affected area for 10–15 minutes, a topical solution such as calcium gluconate 2.5% gel (which has a strong following as the treatment of choice) should be massaged into the affected area. If the original HF solution was less than 20%, the patient may also be given a supply of gel to take with them for repeat applications. If the original solution was stronger than 20% or there was a delay in the rinsing the affected area or treatment, or the patient feels pain after the topical treatment for 45 minutes, a 5–10% calcium gluconate infiltration (injection) may need to be administered.^[9] The bottom line with HF is that prevention of exposure is the best approach with stringent enforcement of

the use of appropriate PPE (including pre-testing all acid gloves for pin-hole leaks prior to use), detailed housekeeping (cleaning up all residues on work surfaces even if they look like water), and constant training of personnel with potential for exposure to this acid. This training should include treating all skin contact with unknown liquids as HF until analysis (such as the use of HF-specific dip sticks) show otherwise.

2.4 Chemically Protective Glove Use Per Process Area.

Chemical-protective (CP) gloves may be required for many of the operations performed in cleanrooms. The primary processing steps that require the use of CP gloves are oxidation, photolithography, etching, junction formation, deposition, and metallization. A brief description of the types of dermal hazards that may be encountered in each semiconductor processing step is presented in Table 5.1.^{[2][5][10]} Much of the focus on CP gloves has focused on the lithography processing area due to concerns regarding the dermal exposure of workers to the glycol ether compounds used in photoresist application and development processes.

Table 5.1. Chemical-Protective Gloves Used Per Processing Step.^{[2][5][10]}

Processing Step	Type of Gloves	Special Hazards
Oxidation		
cleaning	Acid resistant—usually natural rubber, nitrile or latex	Protection against hydrofluoric acid.

Photolithography		
precleaning	Acid resistant—usually natural rubber or latex	Protection against hydrofluoric acid (HF) and other acids/mixes (H ₂ SO ₄ , H ₃ PO ₄ , HCl, HNO ₃ , and H ₂ O ₂).
photoresist application	Solvent resistant—usually nitrile, neoprene, natural rubber; blends; butyl offers the best protection, or Viton® for special applications	Special concern over the permeation of glycol ethers, and the mixtures of xylene, n-butyl acetate, acetone, and ethyl benzene. The use of HMDS in vapor priming operation requires special selection criteria.

Table 5.1. (cont'd)

Processing Step	Type of Gloves	Special Hazards
Photolithography (Cont'd)		
developing	Solvent resistant as above for negative PRs; corrosive resistant for positive developers	As above, except positive PR's utilize a dilute mixture of hydroxides.
stripping: wet chemical	Acid resistant—usually nitrile, natural rubber, or blends; butyl for M-Pyrol	N-Methylpyrrolidone (M-Pyrol) has an unusual characteristic of attacking most polymers and elastomers. Butyl rubber gloves are typically recommended.
stripping: dry chemical	Acid resistant—usually natural rubber, nitrile or blends	Cleaning of etch chambers, traps, and other maintenance activities, corrosive materials—usually acids.
Etching		
wet and dry	as above: stripping	as above: stripping.
Junction Formation		
diffusion	Acid resistant—usually natural rubber, nitrile, or blends; toxic metals: surface protective gloves such as latex or vinyl; spin-on dopants: solvent-type gloves	Highly corrosive liquid sources such as: POCl_3 , BBr_3 , HBr , and BCl_3 are sometimes used, and require the use of extreme precaution to avoid skin contact. Toxic metals such as antimony are used in collector deposition. They are surface-contact hazards and require good housekeeping and personal hygiene. Past use of spin-on dopants that contained a carrier solvent required use of solvent resistant gloves.

Table 5.1. (Cont'd)

Processing Step	Type of Gloves	Special Hazards
Junction Formation (cont'd)		
ion implantation	Acid resistant—usually natural rubber, nitrile, or blends; toxic metals: surface-protective gloves such as latex or vinyl	Highly corrosive gases are used such as BF ₃ , PF ₅ and AsF ₅ which leave highly corrosive residues in source housings, traps, and pump systems. Toxic metals such as arsenic and phosphorus are surface-contact hazards and require good house-keeping and personal hygiene. Phosphorus is a fire hazard.
Deposition		
Epitaxial	Acid resistant—usually natural rubber, nitrile, or blends	Use of highly corrosive compressed liquids or gaseous materials such as HCl, SiCl ₄ , SiH ₂ Cl ₂ and SiHCl ₃ that leave corrosive residues in reactor chambers, gas dispense systems, and pump systems.
CVD	Acid resistant—usually natural rubber, nitrile, or blends	Limited use of corrosive materials such as HCl, SiCl ₄ , and ammonia that leave corrosive residues in reactor chambers, gas dispense systems, and pump systems.
Metallization	Acid resistant—usually natural rubber, nitrile, or blends	Strong acids (fuming nitric and aqua regia) are used for noble metal liquid etching (gold and platinum) and require precautions in the use CP gloves. Dry metal etching uses highly corrosive compressed gases and liquids (as detailed in Epitaxial - above).

Table 5.1. (Cont'd)

Processing Step	Type of Gloves	Special Hazards
Other		
• Maintenance • Spill Response • Hazardous Waste Handling	All of these functions require the use of CP gloves that are the most protective available.	Typically, utilize a double layer of CP gloving (such as a latex interior, and exterior of Viton® or butyl rubber, etc.). Extreme precaution must be taken in the selection of CP gloves and protective garments for these applications

2.5 Glove Selection

The proper selection of gloves or other protective garments that act as a barrier against direct physical contact with the skin or other body surfaces is made difficult by the wide range of hazardous chemicals encountered in semiconductor processing. The prevalence of mixtures of chemicals and manufacturing-specific formulations have permeation characteristics that are usually not scientifically tested or published, and are based only on empirical data from actual use in the industry.

The complexity and variety of chemical operations performed in the semiconductor industry (as detailed in Table 5.1) makes scientifically precise glove selection extremely difficult. Generally, the most hazardous material that may be encountered is used to define minimum CP glove selection, with unusual property materials being factored into any final selection. Even more difficult is defining a set of workable procedures for specific change-out frequencies for gloves based on a combination of skin contact hazard and glove permeation rate. Therefore most semiconductor manufacturers set a generic requirement that all chemical protective gloves are changed when internal contamination of the glove is suspected or at the end of each shift, whichever is sooner. Obviously, at the first sensation of wetness or irritation the glove should be disposed of in a hazardous waste container.

2.6 Pre- and In-Use Glove Testing

Additionally, most semiconductor manufacturers require that employees pretest all CP gloves prior to use. The testing protocol generally specifies the nitrogen inflation test, and the submersion in a DI water bath (looking for bubbles) or tying the glove off and waiting a specified time period for deflation (this method will help to detect minor or *pin-hole* size leaks that have historically been one of the sources of dermal contact burns in the industry).

The rationale for these testing and change requirement's are that when a chemical is absorbed by the CP glove material, the chemical initiates a degradation of the physical integrity of the glove, and materials begin to physically permeate into the matrix of the glove material. Short duration washing of the material with soap and water removes some portion of the surface contamination, but not the absorbed chemical. After surface decontamination, some of the absorbed chemical will continue to permeate the glove material (e.g., overnight) and may ultimately appear on the inside surface.^[5]

The amount of material that permeates to the inside surface after decontamination usually is not enough to be a hazard. A more important concern is that once the material has been penetrated, subsequent contact with the chemical will result in breakthrough times far less than normal. Because the altered breakthrough times can not be calculated, the effectiveness of the glove can not be ensured even when there are no visible signs of a problem, and should be properly discarded.^[10]

3.0 RESPIRATORY PROTECTION

The nature of semiconductor manufacturing helps to minimize chemical exposures during normal manufacturing operations. Also, process equipment manufacturers typically design in additional controls to minimize chemical releases.^[12] Therefore, exposures during production operations are typically far below recommended and legal limits. For this reason, respirator usage within the semiconductor industry is normally confined to emergency responses and some short duration maintenance operations

where other controls are not feasible or additional protection is desired including:

- *Self-contained breathing apparatus* (SCBA) used during emergencies (e.g., major chemical spills, toxic gas releases/air monitoring alarm conditions, or investigation of “unknown odors”)
- SCBA and airline respirators, with an emergency escape bottle, used as a precautionary measure during the changing of toxic gas cylinders (e.g., arsine, phosphine, and diborane)
- Airline respirators or air purifying cartridge respirators used during certain equipment maintenance operations most notably those involving arsenic (e.g., cleaning GaAs crystal growers and cleaning source housings of ion implanters using solid or gaseous arsenic)

Respirators as part of a routine operation should only be considered when other controls are infeasible or insufficient. Key elements of a respirator program include:

- Selection of the proper respirator based on the chemicals, exposure concentrations, and employee preference
- Medical surveillance of the respirator wearer
- Fit testing of the respirator wearer to ensure a proper sizing and face seal
- Training of the respirator wearer on the use and care of the respirator
- Periodic checks on the effectiveness of the program

The basis of a good respirator program are contained in Refs. 13–16. They should be consulted for additional program details.

A comprehensive respirator program involves many regulation-driven requirements and procedures. Some guidance on the implementation of a comprehensive program involves:

1. Evaluate the materials in use, exposure parameters, and IH data to define appropriate equipment.
2. Identify persons who are required to wear respiratory protection, and ensure they are properly trained and maintain current certification.

3. Ensure that every person who uses a respirator is fully informed regarding the hazards of the task they will be performing, and is instructed in the proper use of the device.
4. Ensure employees use and maintain all required respiratory devices in a clean and sanitary condition.
5. Confirm that the person is physically able to perform the work while using a respirator. A local occupational health physician should determine what health and physical conditions are pertinent.

The industrial hygienist at the manufacturing location should:

1. Establish a written respiratory protection program.
2. Conduct air sampling to evaluate the need for respiratory protection.
3. Assist with the selection and procurement of respirators.
4. Develop instructions on the fitting and use of respirators, and ensure employee training is provided and certifications are issued for respirator use.
5. Conduct regular inspections and evaluations to determine the continued effectiveness of the program.

Exposure from breathing air contaminated with harmful dusts, fogs, mists, smokes, sprays, or vapors should be controlled as far as practical by engineering controls, such as enclosures or LEV. When effective engineering controls are not feasible, appropriate respiratory protection procedures must be developed, and protective equipment must be used.

In addition to air-supplied respirators and cartridge respirators, dust masks are used during a variety of operations within the semiconductor industry where typically the chemical exposures are below levels considered significant. Often the dust masks are worn for short duration operations to protect against annoying nuisance dusts present at levels below the occupational exposure limit. Another frequent use for the dust masks is to minimize exposure to toxic dusts present at levels far below allowable limits such as during typical dross removal from a wave solder machine or adding antimony trioxide to a collector deposition weigh-out and load station on a diffusion furnace.

While requirements for the use of dust masks are not as great as other air purifying respirators, they can present a hazard to the user under certain circumstances. Because of this, the following actions need to be taken:

- A decision needs to be made by the site's consulting physician on what, if any, medical surveillance is needed for the users
- Users need to be educated in the use and limitations of the dust masks
- Distribution of the dust mask needs to be controlled to prevent unauthorized use

Also, operations where dust masks are used need to be evaluated and the results of the evaluation should be included with the documentation for the respirator program to show the dust masks are adequate for the task.

When an employee working in a semiconductor operation desires to wear respiratory protection for personal reasons, the job task, environment, and circumstances must be reviewed by the supervisor, department manager, nurse, and designated health and safety person. If a respirator is worn for personal reasons, the wearer should be included in the respiratory protection program. As a minimum, the wearer must receive health surveillance, and training on usage care and limitations of the respirator. Also, if a respirator is worn for personal reasons, the fact should be documented as part of the written respiratory protection program. Some examples of processing locations where nuisance dusts may annoy the operators are:

- Pyrox or Silox operation where the reactor platter may need to be vacuumed for amorphous silicon dioxide residues
- During bead blasting operations where non-contaminated processing equipment parts are being cleaned
- During surface cleaning operations where physical residues of non-hazardous processes create particulate levels, such as cleaning the packing at the load end of diffusion furnaces (assuming the furnace is newer and no asbestos is present)

Industrial hygiene monitoring should be performed for each non-emergency operation or procedure where a respirator is used, including where respirators are worn for personal reasons. Only respirators which are approved and provided by the semiconductor manufacturer, and certified by competent authority (e.g., UK - HSE; U.S. - NIOSH or MSHA) should be allowed to be worn by employees.

Respirator selection should be made according to generally recognized standards (e.g., ANSI Z88.2, *Practices for Respiratory Protection*).^[15]

Whenever respirators are used, a written program which addresses the selection and use of respirators should be in place and include the following elements:

- The name of the individual designated as responsible for implementing and maintaining the respiratory protection program.
- The name of each operation or procedure where respirators are used.
- The type of respirator used for each operation or procedure.
- Conditions under which the respirators are to be used. For each operation or procedure, this should include:
 - (a) Chemical(s)
 - (b) Average exposure concentration
 - (c) Peak exposure concentration

3.1 Training

Users must be thoroughly trained and knowledgeable about the nature of the contaminants, the potential exposure, and the proper use, care, and limitations of the respirator. Refresher training should be performed at an appropriate frequency, usually at one year intervals.

3.2 Fit Testing

At the time of issuance, the respirator must be tested to determine that an effective face seal is possible. Employees not able to demonstrate a good facial seal must not be allowed to wear a respirator. For this reason, employees with beards should not be allowed to wear a respirator for ERT purposes or when airborne chemical concentrations are significant. The use of respirators by bearded employees in non-hazardous environments is a policy decision to be made at each individual site (and considered controversial). Fit tests must be repeated at an appropriate frequency, typically one year.

3.3 Maintenance and Care

Respirators must be cleaned, disinfected, inspected, and properly stored after each use. Any part that is worn or deteriorated must be replaced.

3.4 Health Surveillance

Any employee assigned to a task that requires the use of a respirator must receive a health examination to determine that the person is physically able to perform under such conditions (see Ref. 16). Additionally, evaluations must be made periodically to determine that medical conditions affecting the safe use of respirators have not changed (see Ref. 16).

3.5 Documentation

Records should be kept of the following:

- Inspection dates and findings for respirators maintained for emergency use
- Industrial hygiene monitoring, relative to the respiratory protection program
- Health surveillance results
- Job assignment information for users
- Training curricula and attendance records
- Maintenance and repair records

For U.S. based operations, industrial hygiene monitoring records must be maintained in accordance with OSHA regulations (30 year minimum); health surveillance results must be maintained in accordance with OSHA regulations; all other documentation for the respiratory protection program should be maintained for a minimum of five years. See Ch. 8: IH Recodkeeping, for additional details.

3.6 Program Evaluation

An annual evaluation should be conducted to determine whether respirators are being correctly used, maintained, and stored. This evaluation should include physically inspecting a representative sample of the respirators used on the site. The representative sample should include at least 10 percent of the respirators used on site.

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